

National Technical University of Athens

School of Electrical and
Computer Engineering (ECE)

10th semester

Control Systems of Electrical Machines

Simulation Assignment 3: Induction Motor Vector Control, Current Model

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Induction Motor Rated Quantities

Machine Parameters

An induction motor rated at 2.3 kW is used for the modelling. The machine parameters are presented in Table (1):

Quantity	Symbol	Value (SI)	Value (p.u.)
Rated power	P_n	2.3 kW	1
Rated voltage	V_n	230 V	1
Rated current	I_n	3.6 A	1
Frequency	f	50 Hz	1
Pole pairs	n_p	2	–
Synchronous speed	n_s	1500 rpm	1
Rated speed	n_n	1410 rpm	0.94
Slip	s	0.06	0.06
Stator resistance	R_s	6.5746 Ω	0.457
Rotor resistance	R_R	2.106 Ω	0.146
Magnetizing inductance	L_M	0.3354 H	2.95
Leakage inductance	L_σ	0.0416 H	0.366
Moment of inertia	J	0.01 kg·m^2	–
Damping coefficient	b	0	–

Table 1: Parameters of the IM machine

Per-Unit Bases

The bases used to convert the system to per-unit quantities are presented below. The machine phase voltage and phase current are selected as bases. A normalization factor $K = 1$ is used, so the quantities are normalized with respect to their peak values.

$$S_{base} = 2.3 \text{ kVA} \tag{1}$$

$$V_{base} = \sqrt{2}KV_n = \sqrt{2} \cdot 230 = 325.27 \text{ V} \tag{2}$$

$$I_{base} = \sqrt{2}KI_n = \sqrt{2} \cdot 3.6 = 5.09 \text{ A} \tag{3}$$

$$Z_{base} = \frac{V_{base}}{I_{base}} = 63.89 \ \Omega \tag{4}$$

$$\omega_{base} = 2\pi f = 314.16 \text{ rad/s} \tag{5}$$

$$\psi_{base} = \frac{V_{base}}{\omega_{base}} = 1.0355 \text{ Wb} \tag{6}$$

$$T_{base} = \frac{3n_p V_{base} I_{base}}{2\omega_{base} K^2} = 14.64 \text{ Nm} \tag{7}$$

The rated rotor flux is calculated from:

$$\psi_N^R = \sqrt{\frac{1 - L_{\sigma,pu}^2}{1 + 2L_{\sigma,pu}/L_{M,pu}}} \cdot \psi_{base} = 0.867 \cdot 1.0354 = 0.9072 \text{ Wb} \tag{8}$$

The rated electromagnetic torque of the machine is calculated as:

$$T_{e,N} = \frac{3n_p}{2K^2}\frac{V_{base}I_{base}}{\omega_{base}}\frac{1}{1+2L_\sigma/L_M}\sqrt{\left(1-L_{\sigma,n}^2\right)\left[\left(1+\frac{L_\sigma}{L_M}\right)^2-\frac{1}{L_{M,n}^2}\right]} = 11.734 \text{ Nm} \tag{9}$$

1 Phase-Locked Loop (PLL)

In this question, a PLL controller was implemented to estimate the rotor-field orientation angle θ_e, with the aim of aligning the d axis with the rotor-flux vector. Its operation is based on minimizing the q component of the flux in the rotating reference frame.

In this case, direct measurement of the rotor field is used, namely the components ψ_R^α and ψ_R^β from the machine model.

The implemented PLL uses the following equations to calculate the estimated frequency and field-orientation angle:

$$\omega_1 = \frac{\alpha_l}{p+\alpha_l}\left(\widehat{\omega}_g - k_{pp}\psi_q - k_{ip}\int \psi_q\, dt\right) \tag{10}$$

$$\theta_1 = \text{mod}\left(\int \omega_1\, dt, 2\pi\right) \tag{11}$$

where:

- $\widehat{\omega}_g$: the initial estimate of the PLL output frequency, set to zero in this case.
- α_l: the low-pass filter cutoff frequency.

The PLL parameters were selected from the following relationships, assuming a current-controller rise time of $t_{rc} = 1$ ms:

$$a_c = \frac{\ln(9)}{t_{rc}} = 2197.2 \text{ rad/s}, \qquad a_p = \frac{a_c}{20} = 109.86 \text{ rad/s}, \qquad a_l = 10a_p = 1098.6 \text{ rad/s} \tag{12}$$

The gains of the PLL PI controller are therefore selected as:

$$k_{pp} = \frac{2a_p}{\psi_N} = 242.204, \qquad k_{ip} = \frac{a_p^2}{\psi_N} = 13304.421 \tag{13}$$

Figure (1) shows the implemented model, which includes scalar speed control developed in the previous simulation assignment. The PLL-controller subsystem is shown in Figure (2).

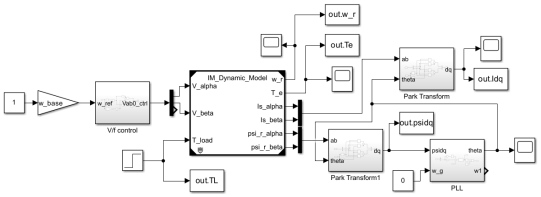


Figure 1: IM model

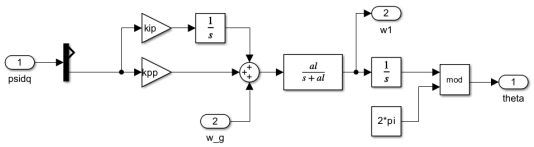


Figure 2: PLL sub-system

Park Transformation

The Park transformation converts quantities from the stationary $\alpha\beta$ reference frame to a rotating dq coordinate system that rotates synchronously with the rotor field. This decouples flux production from torque production, greatly facilitating vector control of the machine.

The transformation is described by Equation (14), where θ_e is the electrical orientation angle of the rotating reference frame, calculated by the PLL:

$$\begin{bmatrix} x_d(t) \\ x_q(t) \end{bmatrix} = \begin{bmatrix} \cos\theta_e & \sin\theta_e \\ -\sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} x_\alpha(t) \\ x_\beta(t) \end{bmatrix} \tag{14}$$

Similarly, the inverse Park transformation converts quantities from the rotating dq frame back to the stationary $\alpha\beta$ frame and is given by:

$$\begin{bmatrix} x_\alpha(t) \\ x_\beta(t) \end{bmatrix} = \begin{bmatrix} \cos\theta_e & -\sin\theta_e \\ \sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} x_d(t) \\ x_q(t) \end{bmatrix} \tag{15}$$

Figures (3) and (4) show the Park and inverse Park transformation subsystems, respectively.

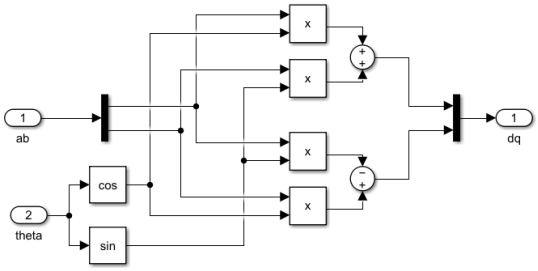


Figure 3: Park transform sub-system

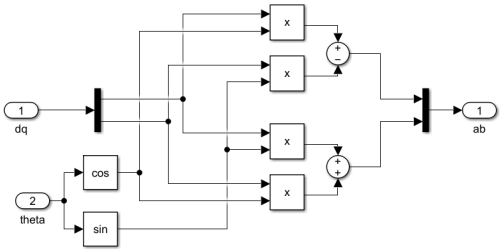


Figure 4: Inverse Park transform sub-system

Simulation Results

Figures (5)–(8) show the speed, electromagnetic and load torque, dq current components and corresponding rotor-flux components in the rotating reference frame, expressed in per-unit values. Torque is normalized with respect to the rated torque $T_{e,N} = 11.734$ Nm, rather than the base torque.

In this simulation, a transition from no load to half-rated load torque is applied at 0.7 s.

The results show a pronounced starting transient in both torque and stator currents, after which the system settles into steady-state operation.

Rotor speed maintains its rated value in steady state, with a small deviation after the transient because of the motor slip required to produce half the rated torque.

Regarding the current components, both the currents and the flux in the dq frame oscillate strongly during the first 0.15 s of the simulation. This occurs because the PLL has not yet locked onto the correct angle.

In steady state, $i_d = 0.538 \cdot 5.09 = 2.74\ A$ remains constant because it produces the rotor flux, while i_q varies with load torque. Thus, after the machine accelerates to rated speed, i_q becomes zero under no load. After 0.7 s, it maintains a constant positive value of $i_q = 0.437 \cdot 5.09 = 2.225$ to compensate for the applied load torque.

This behaviour can also be verified quantitatively using the theoretical field-orientation relationships, in steady state and with full rotor-field alignment. The expected magnetizing-current value is:

$$i_d = \frac{\psi_R}{L_M} = \frac{\psi_N}{L_M} = \frac{0.905\ \text{Wb}}{0.3354\ \text{H}} = 2.70\ \text{A} \tag{16}$$

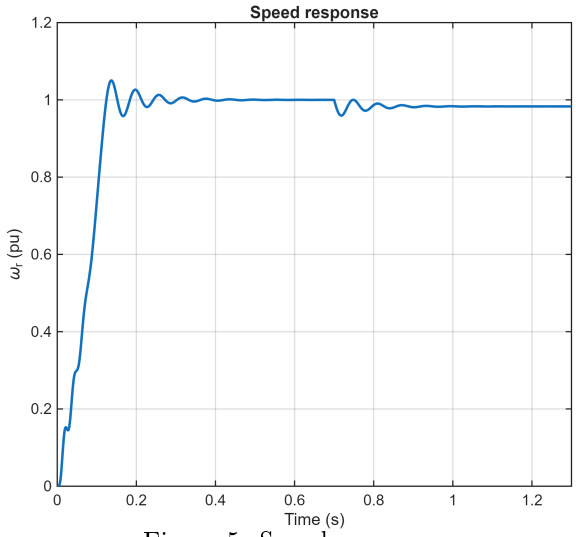


Figure 5: Speed response

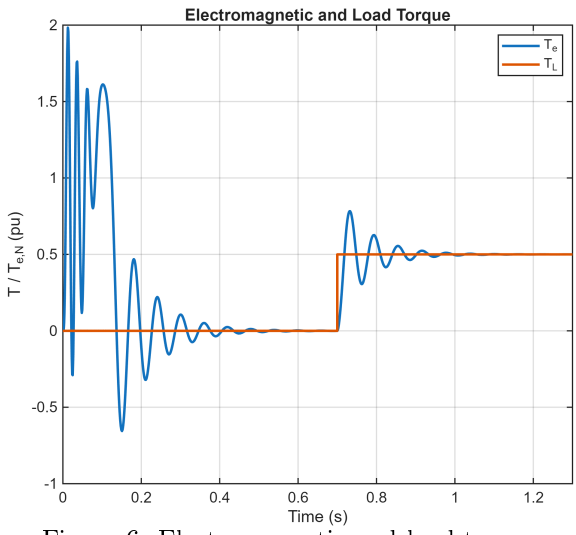


Figure 6: Electromagnetic and load torque

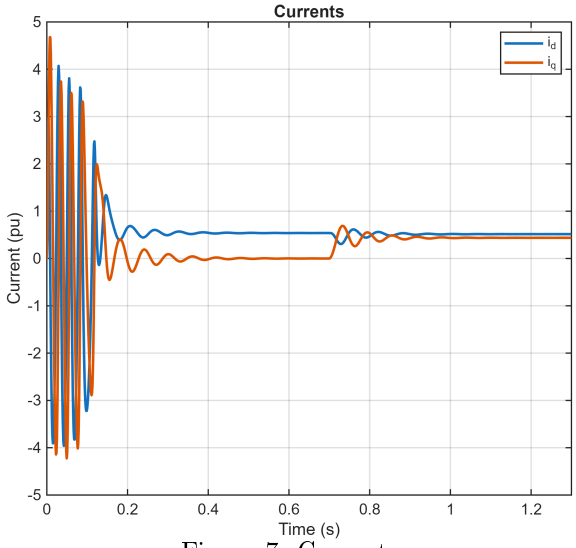


Figure 7: Currents

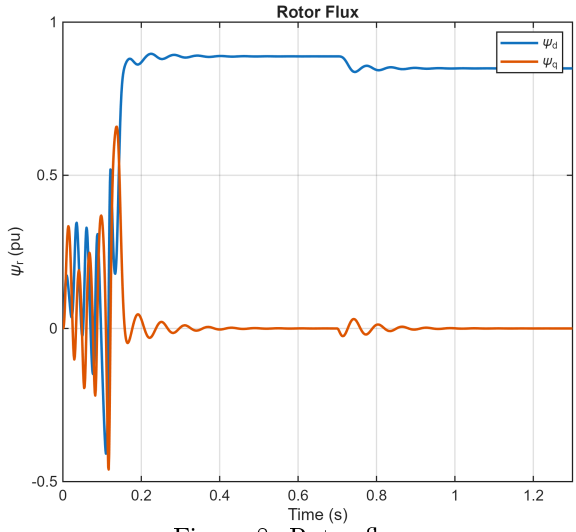


Figure 8: Rotor flux

Similarly, the electromagnetic torque satisfies:

$$T_{e,N} = \frac{3}{2K^2} n_p \psi_N i_q \tag{17}$$

Therefore, for a load equal to half the rated torque:

$$i_q = \frac{2K^2(T_{e,N}/2)}{3 n_p \psi_N} = \frac{2 \cdot 1^2 \cdot (11.734/2)}{3 \cdot 2 \cdot 0.905} = 2.16 \text{ A} \tag{18}$$

The theoretical calculations agree satisfactorily with the theoretical predictions. This comparison provides additional confirmation of correct field-orientation and PLL operation.

Regarding the rotor-flux components in the dq frame, ψ_d remains almost constant, while ψ_q stays very close to zero. This confirms successful alignment of the d axis with the rotor field and, consequently, correct operation of the PLL and field orientation.

2 Current Control in the Rotating Reference Frame

In this question, a two-degree-of-freedom current controller was implemented in the rotating dq reference frame to control the stator-current components i_d and i_q independently. The current controller is described by the following equations:

$$\begin{bmatrix} e_d \\ e_q \end{bmatrix} = \begin{bmatrix} i_d^{ref} \\ i_q^{ref} \end{bmatrix} - \begin{bmatrix} i_d \\ i_q \end{bmatrix} \tag{19}$$

$$\begin{bmatrix} v_d^{ref} \\ v_q^{ref} \end{bmatrix} = k_p \begin{bmatrix} e_d \\ e_q \end{bmatrix} + k_i \frac{1}{s} \begin{bmatrix} e_d \\ e_q \end{bmatrix} - \begin{bmatrix} R_a & \omega_1 L_\sigma \\ -\omega_1 L_\sigma & R_a \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} E_d \\ E_q \end{bmatrix} \tag{20}$$

The physical meaning of each controller equation is explained below:

- Current error (19): The current-error vector is the difference between the desired currents (i_d^{ref}, i_q^{ref}) and the actual currents (i_d, i_q). The controller aims to eliminate the error on both axes so that the actual currents follow their references.
- PI controller with decoupling and feedforward (20): This equation describes a two-degree-of-freedom PI controller in the dq reference frame. The first two terms represent proportional and integral action, ensuring a fast response and zero steady-state error, respectively.

  The controller also includes three additional compensation terms:

  - The R_a term introduces active resistance, increasing system damping.
  - The $\pm\omega_1 L_\sigma$ term compensates for cross-coupling caused by rotation of the reference frame. This allows independent control of the d and q axes and simplifies controller design.
  - The E_d and E_q terms provide feedforward compensation of the machine back EMF, given by:

$$\hat{E} = j\omega_1 \hat{\psi}_R \tag{21}$$

The controller gains and active-resistance value are shown below, assuming a rise time of $t_{rc} = 1$ ms:

$$k_p = \alpha_c L_\sigma = 91.405 \qquad k_i = \alpha_c^2 L_\sigma = 200840 \qquad R_a = \alpha_c L_\sigma - R_s = 82.724 \tag{22}$$

Figure (9) shows the implemented model, while Figure (10) shows the current-controller subsystem.

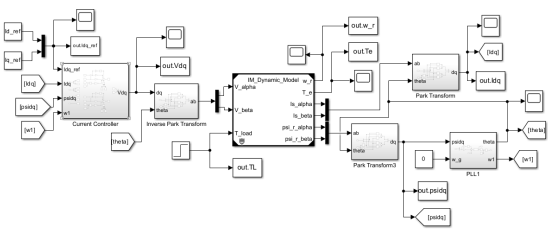


Figure 9: IM model

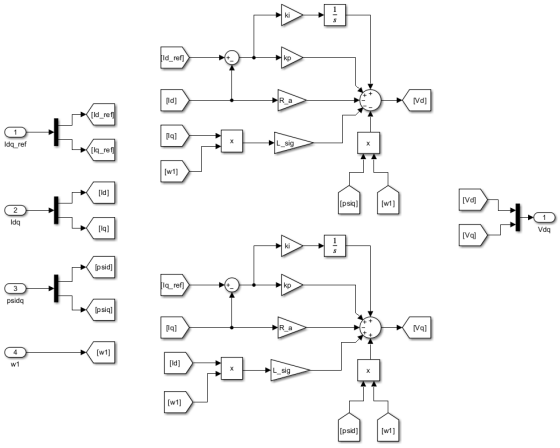


Figure 10: Current controller

Simulation Results

Figures (12)–(14) show the current and voltage components in the rotating dq reference frame, together with the current responses used to verify the controller rise time.

In this case, motor speed is not controlled directly; current references are imposed directly in the rotating reference frame. They are selected so that the machine operates at its rated quantities, namely rated rotor flux and rated electromagnetic torque.

$$\psi_N = L_M i_d^{ref} \implies i_d^{ref} = \frac{\psi_N}{L_M} = 2.705 \text{ A} = 0.5313 \text{ p.u.} \tag{23}$$

$$T_{e,N} = \frac{3}{2K^2} n_p \psi_N i_q^{ref} \implies i_q^{ref} = \frac{T_{e,N}}{\frac{3}{2K^2} n_p \psi_N} = 4.313 \text{ A} = 0.8472 \text{ p.u.} \tag{24}$$

During the simulation, load torque changes from 0 to 80% of its rated value at 0.3 s.

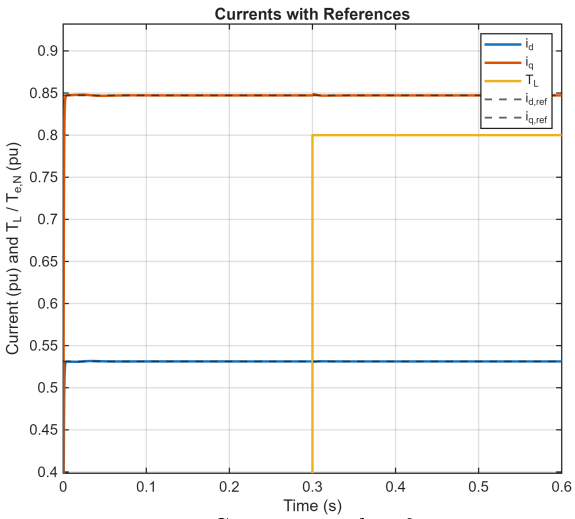


Figure 11: Currents with references

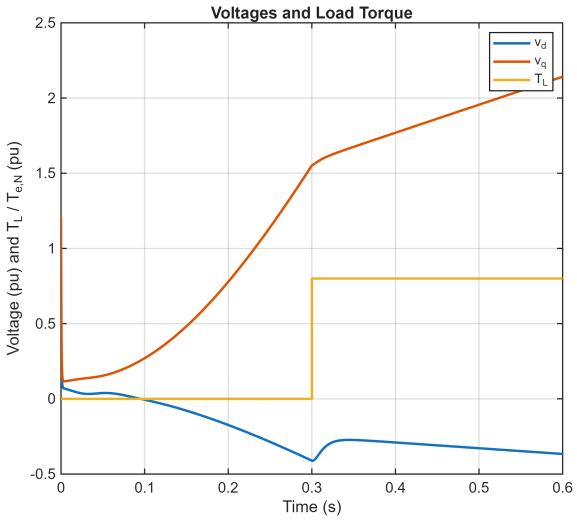


Figure 12: Voltages and load torque

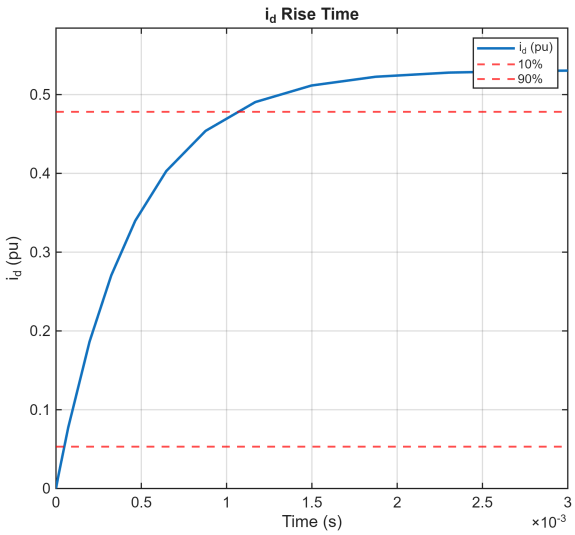


Figure 13: D-axis current rise time

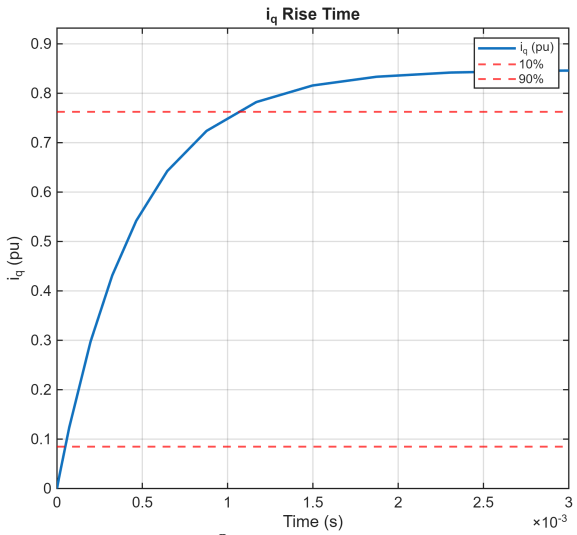


Figure 14: Q-axus current rise time

The results show that the current components i_d and i_q follow their respective references very accurately, without being affected by the load change. This confirms correct current-controller operation in the rotating reference frame.

The voltages v_d and v_q also vary dynamically and exceed the machine's rated values because appropriate saturation has not yet been added at the current-controller output.

Finally, the current responses in Figures (13) and (14) confirm the controller rise time t_{rc}, defined as the time required for the response to increase from 10% to 90% of its final value:

$$t_{rc,d} = t_{rc,q} = t_{90\%} - t_{10\%} \approx 1.162 - 0.0714 = 1.0906\ msec \tag{25}$$

3 Voltage Saturation and Anti-Windup

In this question, the rotating-frame current controller was extended to include voltage-reference limiting and anti-windup protection for the PI-controller integral terms. Specifically, the current-controller equations become:

$$\begin{bmatrix} e_d \\ e_q \end{bmatrix} = \begin{bmatrix} i_d^{ref} \\ i_q^{ref} \end{bmatrix} - \begin{bmatrix} i_d \\ i_q \end{bmatrix} \tag{26}$$

$$\begin{bmatrix} v_d^{ref} \\ v_q^{ref} \end{bmatrix} = k_p \begin{bmatrix} e_d \\ e_q \end{bmatrix} + k_i \frac{1}{s} \begin{bmatrix} e_d + \frac{1}{k_p}(\bar{v}_d^{ref} - v_d^{ref}) \\ e_q + \frac{1}{k_p}(\bar{v}_q^{ref} - v_q^{ref}) \end{bmatrix} - \begin{bmatrix} R_a & \omega_1 L_\sigma \\ -\omega_1 L_\sigma & R_a \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} E_d \\ E_q \end{bmatrix} \tag{27}$$

$$\begin{bmatrix} \bar{v}_d^{ref} \\ \bar{v}_q^{ref} \end{bmatrix} = \frac{V_{base}}{\max(|v_{dq}^{ref}|, V_{base})} \begin{bmatrix} v_d^{ref} \\ v_q^{ref} \end{bmatrix} \tag{28}$$

$$|v_{dq}^{ref}| = \sqrt{(v_d^{ref})^2 + (v_q^{ref})^2} \tag{29}$$

The limiter aims to keep the voltage-reference magnitude within the physical limits of the machine and inverter.

Saturation is implemented by limiting the overall voltage-vector magnitude rather than limiting the two components independently. The voltage reduction is therefore distributed proportionally between the d and q axes, preserving the direction of the voltage-reference vector.

To prevent *integrator windup*, back-calculation anti-windup was used: the difference between the saturated and unsaturated voltages is fed back to the PI-controller integrator.

The new current-controller model is shown in Figure (15).

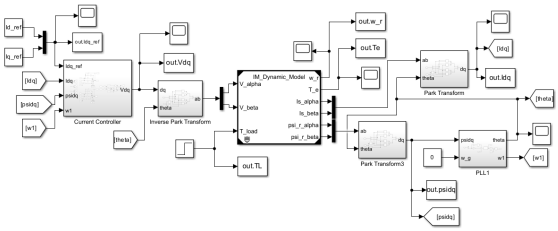


Figure 15: Current controller with saturation

Simulation Results

Figures (16) and (17) show the current and voltage components in the rotating reference frame, together with the load-torque change.

During the simulation, load torque changes to 80% of its rated value at 0.5 s. The current references remain the same as in Question (3), as given by Equations (23) and (24).

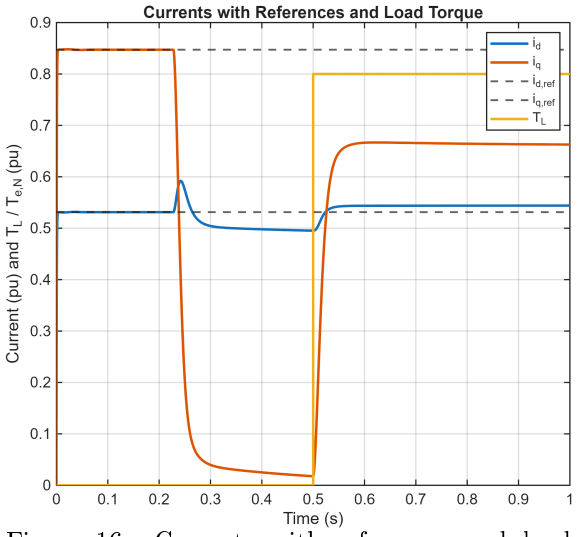


Figure 16: Currents with references and load torque

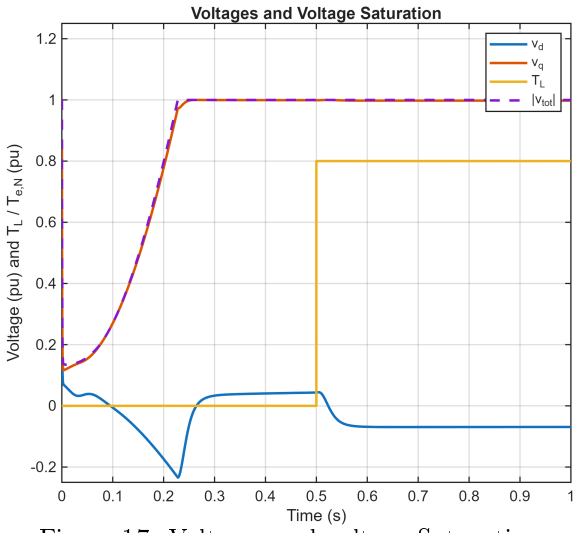


Figure 17: Voltages and voltage Saturation

The results show that the voltage-reference magnitude is successfully limited to the rated value, without overshoot caused by excessive accumulation of integral error.

When the motor reaches rated speed, at approximately 0.2 s, the current controller is constrained by voltage saturation because of the large back EMF. As a result, the current ceases to follow its reference.

Nevertheless, when load is added, the actual current i_q increases to raise electromagnetic torque and compensate for the load.

4 Current Model and Indirect Field Orientation

In this question, a current model was implemented to estimate the rotor-flux vector, replacing the direct flux measurement used in the preceding questions. The flux estimate is used for *Indirect Field Orientation (IFO)* and to calculate the orientation angle of the rotating reference frame.

The current model is based on the rotor equations in the dq reference frame, where rotor flux is calculated from stator currents and slip frequency, as shown by the following expressions:

$$\omega_1 = \omega_r + \frac{\widehat{R}_R i_q^{\text{ref}}}{\widehat{\psi}_R} \tag{30}$$

$$\theta_1 = \text{mod}\left(\int \omega_1 \, dt, 2\pi\right) \tag{31}$$

$$\widehat{\psi}_R = \int \widehat{R}_R \left(i_d^{\text{ref}} - \frac{\widehat{\psi}_R}{\widehat{L}_M}\right) dt \tag{32}$$

Figure (18) shows the overall system, while Figure (19) shows the current-model subsystem.

Simulation Results

Figure (20) shows the estimated rotor-flux components in the rotating dq reference frame together with the actual components, i.e. the machine-model fluxes after the Park transformation. Figure (21)

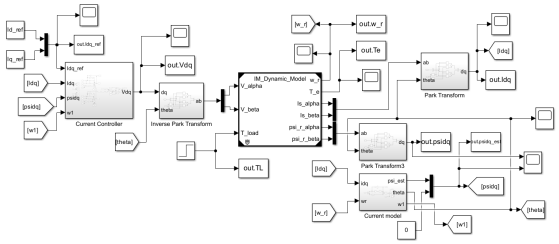


Figure 18: IM model

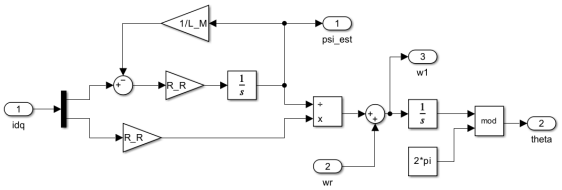


Figure 19: IFO Current model subsystem

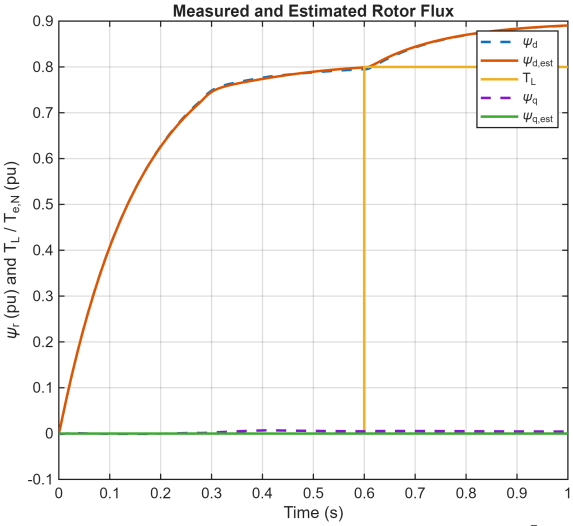


Figure 20: Measured and estimated rotor flux with load torque

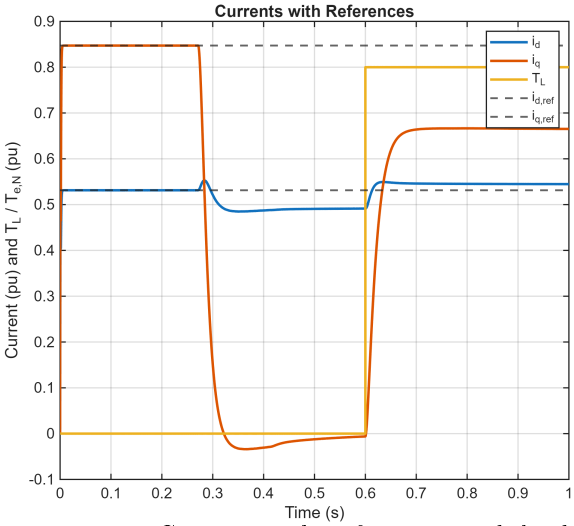


Figure 21: Current with references and load torque

shows the current components and their references, which retain the rated values given by Equations (23) and (24). Load torque changes to 80% of its rated value at 0.6 s.

The results show that the estimated flux components practically coincide with the actual machine-model components, while ψ_q remains zero. This confirms correct operation of the current model and indirect field orientation.

The currents i_d and i_q also behave similarly to the preceding case in Question (3), following their references at all stages except when the current controller enters saturation.

In general, i_d^{ref} and i_q^{ref} are preferred to measured currents to avoid introducing noise into the flux estimator. This was not possible in the present implementation because, when the current controller enters saturation, the current references no longer match the measured quantities.

5 Speed Control

In the final stage of the assignment, a closed speed-control loop was implemented for the induction motor using a two-degree-of-freedom PI controller in the rotating dq reference frame. The controller generates the i_q reference, while the i_d reference remains constant to ensure rated rotor flux, as in the preceding questions.

The speed-controller rise time is selected to be ten times the current-controller rise time:

$$a_s = \frac{a_c}{10} = 219.7225 \text{ rad/s} \tag{33}$$

The controller gains are obtained from:

$$b_a = \frac{2K^2(a_s J - b)}{3n_p^2 \psi_{nom}} = 0.404 \tag{34}$$

$$k_{ps} = \frac{2K^2 a_s J}{3n_p^2 \psi_{nom}} = 0.404 \tag{35}$$

$$k_{is} = \frac{2K^2 a_s^2 J}{3n_p^2 \psi_{nom}} = 88.696 \tag{36}$$

The speed controller is described by:

$$e_s = \omega_{ref} - \omega_r \tag{37}$$

$$i_{q,nom}^{ref} = k_{ps} e_s + k_{is} I_s - b_a \omega_r \tag{38}$$

$$i_q^{ref} = \text{sat}\left(i_{q,nom}^{ref}, \sqrt{I_{base}^2 - (i_d^{ref})^2} \right) \tag{39}$$

$$I_s = \frac{1}{s}\left(e_s + \frac{1}{k_{ps}} (i_q^{ref} - i_{q,nom}^{ref}) \right) \tag{40}$$

Saturation of i_q^{ref} ensures that the total current magnitude remains within the machine limits. The back-calculation term protects the integrator against error accumulation (anti-windup), while b_a increases system damping to improve disturbance rejection.

Figure (22) shows the overall system, and Figure (23) shows the speed-controller model. Since the currents no longer exceed their rated values, the current model is supplied with the references i_d^{ref} and i_q^{ref}.

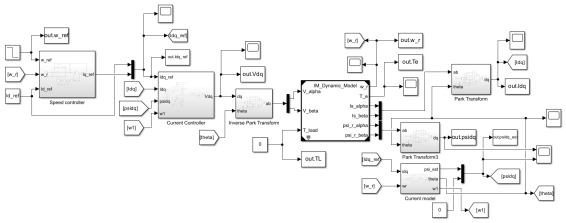


Figure 22: Complete IM model

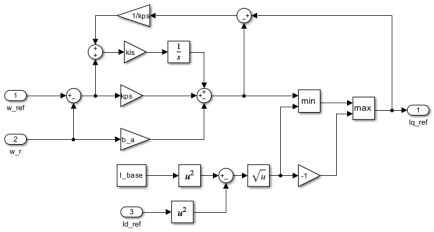


Figure 23: Speed controller sub-system

Simulation Results

To test speed-controller operation, a step change in the speed reference from 1 *p.u.* to 0.5 *p.u.* was applied at 0.5 s, with zero load throughout.

Figures (24) and (25) show the speed response and dq currents together with their respective references.

The speed controller successfully follows the reference change without oscillations, while i_q varies dynamically to produce the electromagnetic torque required for the requested speed changes. In contrast, i_d remains constant, maintaining constant rotor flux.

Figures (26) and (27) show the system rise time with and without saturation, respectively.

With saturation enabled, the rise time increases significantly because the available current, and hence the electromagnetic torque, is limited. Without current limiting, the controller achieves a faster response, approaching the theoretically calculated rise time.

$$t_{r,s}^{sat} = t_{90\%} - t_{10\%} \approx 0.2465 - 0.07131 = 175.2\ msec \tag{41}$$

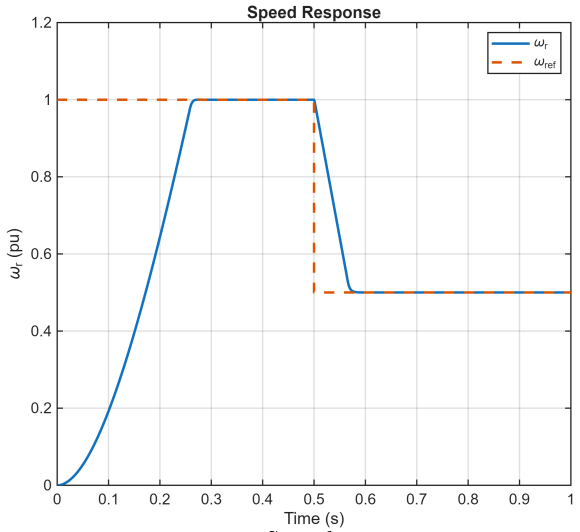


Figure 24: Speed response

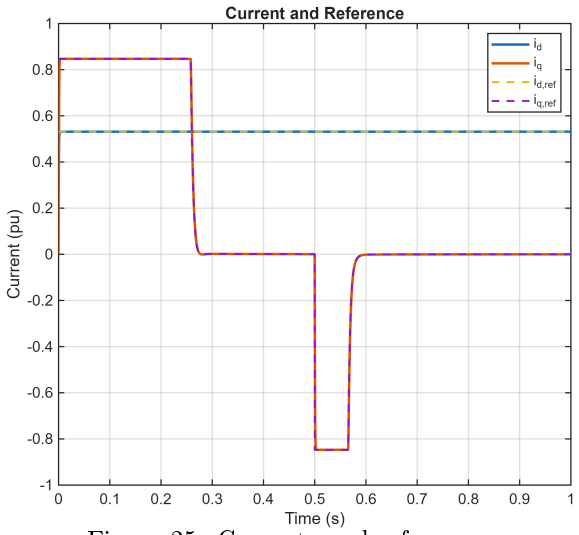


Figure 25: Currents and references

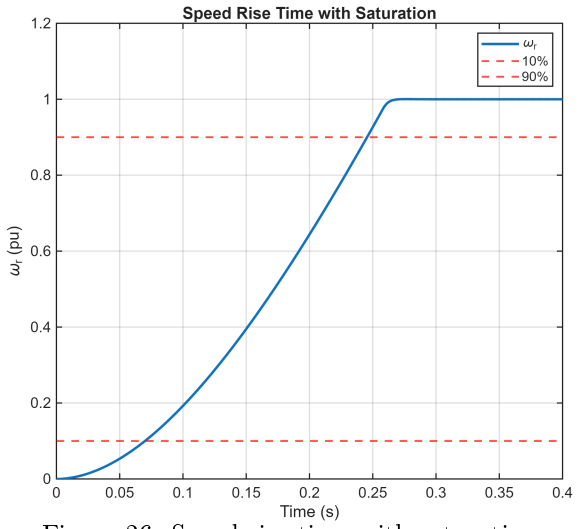


Figure 26: Speed rise time with saturation

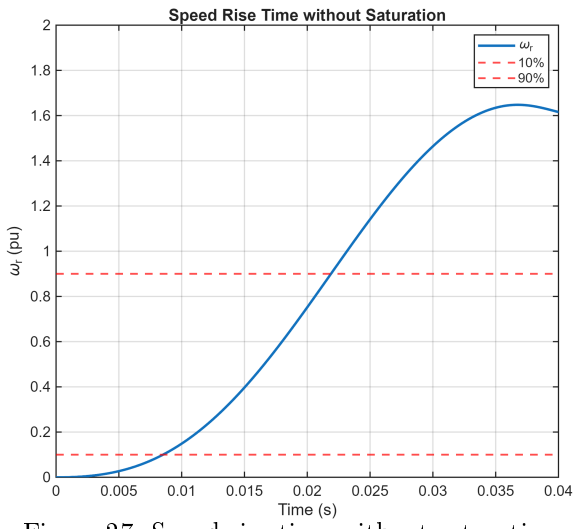


Figure 27: Speed rise time without saturation

$$t_{r,s} = t_{90\%} - t_{10\%} \approx 22.21 - 8.78 = 13.4300\ msec \tag{42}$$

Noticeable speed overshoot occurs without current limiting. This behaviour is not due solely to the absence of saturation; it is also related to the current model used for indirect field orientation. The current model relies on rotor-flux dynamics, characterized by the time constant $\tau = \frac{L_M}{R_M} = 0.159$, so the estimated flux cannot follow current changes instantaneously during a transient. Temporary errors therefore appear in the estimated flux and orientation angle, temporarily reducing the accuracy of decoupling between the d and q axes. As a result, the actual electromagnetic torque momentarily deviates from the value predicted by the controller design, causing speed overshoot. This effect becomes more pronounced when the current limit is removed, since the machine accelerates significantly faster and larger currents develop during the transient. The faster speed variation and increased transient currents amplify the influence of flux- and angle-estimation errors, making the speed overshoot apparent.